Sahaj Bhadja

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Summary

Software Engineer pursuing an MSCS at USC with 2+ years building **end-to-end features**, **search and data systems**, and maintainable applications using **Java**, **Python**, TypeScript, SQL, testing, and CI/CD.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *LA, California*

- Architected a multi-tenant platform using Azure SQL, Blob Storage, row-level security, and normalized schemas for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through query optimization.
- Built **data integration pipelines** using FastAPI, Supabase, SQL Server, and cloud storage, processing 5K+ monthly jobs with validation, reconciliation, error handling, CI/CD releases, and 99%+ production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *LA, California*

- Implemented a production backend on AWS serving 150+ instructors and 10K+ daily requests, using automated deployments and rubric workflows to reduce feedback turnaround 60% while maintaining stable peak-load performance.
- Designed concurrent **REST APIs** with Python multithreading, AWS RDS, and secure file ingestion, processing 1K+ files per batch through fault-tolerant execution, validation, and reliable customer-facing grading workflows.

USC AD Labs **January 2025 – May 2025**
ML Research Assistant *LA, California*

- Built a Python pipeline processing 100K+ trajectory logs, improving long-horizon planning quality 18% through reproducible data preparation, model training, simulation evaluation, automated testing, and detailed experiment analysis.
- Architected a spatial-temporal encoder integrating map geometry and historical agent trajectories through multi-head attention, improving closed-loop planning stability 15% while validating model behavior across complex autonomous-driving scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) *India*

- Designed **OOP modules** with Singleton and Factory patterns for complex API integrations, improving response times 30%, increasing pipeline efficiency 68%, and reducing technical debt 20% through maintainable abstractions.
- Engineered cross-platform microservices with automated builds and 20+ **unit and integration tests** using Jenkins, PyTest, and Mockito, supporting reliable releases across three operating systems within Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) *India*

- Developed a MERN platform with WebSockets, REST APIs, and eight microservices, synchronizing 2M+ records across authentication, billing, permissions, and notification workflows while reducing real-time data conflicts 33%.
- Automated Docker packaging and Jenkins **CI/CD** deployments, cutting release time 50%, while Azure SQL and analytics insights reduced resource waste 42% and increased customer lead conversion 15%.

Skills

Languages: Java, Python, TypeScript, JavaScript, C++, SQL, C#, JSON
Computer Science: Algorithms, Data Structures, OOP, Design Patterns, System Design, Search, Indexing, Data Modeling
Backend & Data: REST APIs, FastAPI, Flask, Node.js, Spring Boot, PostgreSQL, MySQL, SQL Server, MongoDB, Redis, Microservices
Cloud & Engineering: AWS, Azure, GCP, Docker, Kubernetes, Linux, Git, Unit Testing, Integration Testing, PyTest, Mockito, Jenkins, CI/CD, Debugging, Documentation, Agile/Scrum

Projects

EvidenceScope | Python, Search, Algorithms, Legal Data

- Built and tested an **explainable evidence search engine** with inverted indexing, BM25-style ranking, metadata filters, SHA-256 duplicate suppression, deterministic ordering, and matched-term explanations across heterogeneous legal-document records.

Chat2SQL Code | Java, TypeScript, SQL, Docker

- Worked across a **Java** and TypeScript codebase with Docker, Maven, Node.js, and Yarn, supporting 10+ relational and NoSQL databases through schema exploration, query authoring, and cross-platform workflows.

Big-Brain Live | GCP, PostgreSQL, FAISS, REST APIs

- Built a semantic document retrieval system over 50K+ documents using transformer embeddings and FAISS, enabling contextual search and summarization with sub-second retrieval through optimized indexing and similarity pipelines.